

ΠACRO L3 - 2014-2015

Lezione 5

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Session 5

These notes are those I wrote during class. They are posted "for the record".

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Model Structure

period 0 (beginning of time)

[Population L_0 is given
Technologies A_0, B_0 given
given (A_0, B_0, L_0) → optimal behavior
→ market clearing
 $= \lambda(n_0, y_0^*, y_0^*)$

$$(n_0, y_0^A, y_0^N)$$

↳

$$L_1 = L_0 \times n_0$$

$$A_1 = A_0 + y_0^A \quad (\text{learning by doing})$$

$$\Pi_1 = \Pi_0 + y_0^N \quad (\text{L-B-D})$$

period 1

(L_1, A_1, Π_1) given $\rightarrow$ find n_1, y_1^A, y_1^N
 $\rightarrow$ get $L_2, A_2, \Pi_2 \dots$

Let us take a generic period t
 (L_t, A_t, Ω_t) given

I need to compute $\begin{pmatrix} n_t \\ y_t^n \\ y_t^n \end{pmatrix}$

Equilibrium on the food market

$$\left\{ \begin{array}{l} \text{demand: } n_t L_t \\ \text{supply: } Y_t^A = A_t^\varepsilon L_t^{A\alpha} \end{array} \right.$$

$$\text{equil: } n_t L_t = A_t^\varepsilon L_t^{A\alpha}$$

let $\theta_t = \frac{L_t^A}{L_t}$ = share of agricult.
workers

$$\Leftrightarrow L_t^A = \theta_t L_t$$

$$n_t L_t = A_t^\varepsilon (\theta_t L_t)^\alpha$$

$$n_t L_t^{1-\alpha} A_t^{-\varepsilon} = \theta_t^\alpha$$

$$\Rightarrow \theta_t = \left(\frac{n_t L_t^{1-\alpha}}{A_t^\varepsilon} \right)^{\frac{1}{\alpha}}$$

Equl. on the labor market

(note: eq. on the good market by
Walras law)

eg: $w_t^A = w_t^N$

and $w_t^A = \frac{p_t y_t^A}{L_t^A}$, $w_t^N = \frac{1 \times y_t^N}{L_t^N}$

$$w_t^A = w_t^N$$

$$\Leftrightarrow p_t \frac{y_t^A}{L_t^A} = \frac{y_t^N}{L_t^N} \Leftrightarrow p_t \frac{A_t^\varepsilon L_t^{A^{1-\alpha}}}{L_t^A} = \frac{n_t^\varepsilon L_t^N}{L_t^N}$$

$$\Leftrightarrow p_t A_t^\varepsilon (\theta_t L_t)^{1-\alpha} = n_t^\varepsilon$$

$$L_t^A = \theta_t L_t$$

$$n_t = \frac{\gamma}{R_t}$$

$$\Leftrightarrow (*) p_t = \frac{n_t^\varepsilon}{A_t^\varepsilon} (\theta_t L_t)^{1-\alpha}$$

2 unknowns:
 p_t, θ_t

I know that

$$P_t = \frac{\alpha}{n_t}$$

and

$$\theta_t = \left[\frac{n_t L_t^{1-\alpha}}{A_t^\varepsilon} \right]^{\frac{1}{\alpha}}$$

therefore (*) becomes

$$\frac{\alpha}{n_t} = \frac{n_t^\varepsilon}{A_t^\varepsilon} \times L_t^{1-\alpha} \times \left[\frac{n_t L_t^{1-\alpha}}{A_t^\varepsilon} \right]^\alpha \quad (*)$$

n_t is the only unknown.

(★) $n_t = \gamma \frac{A_t^\varepsilon}{n_t^\alpha \varepsilon} \frac{1}{(\gamma L_t)^{1-\alpha}}$

SOLUTION for n_t

Now

$$p_t = \frac{\gamma}{n_t} = \frac{n_t^\alpha \varepsilon}{A_t^\varepsilon (\gamma L_t)^{1-\alpha}}$$

$$\theta_t = \left(\frac{n_t}{A_t^\varepsilon} L_t^{1-\alpha} \right)^{\frac{1}{\alpha}} \varepsilon$$

$$\rightarrow L_t^A = \theta_t L_t \rightarrow y_t^A = \Lambda_t^E (\theta_t L_t)$$

$$\rightarrow L_t^D = (1 - \theta_t) L_t \rightarrow y_t^D = M_t^E L_t^D$$

and then

$$L_{t+1} = n_t L_t$$

$$K_{t+1} = A_{t+1} y_t^A$$

$$n_{t+1} = n_t + y_t^D$$

Where does the economy converge to?

• Balanced Growth Path ?

• Steady state ?

→ $g_t^X =$ growth rate of X
 $= \frac{X_{t+1} - X_t}{X_t}$

$$g_t^A = \frac{y_t^A}{A_t}$$

$$(LB-D) = A_t^{\varepsilon-1} (L_t^A)^{\alpha}$$

$$g_t^n = \frac{y_t^n}{n_t} = n_t^{\tau-1} z_t^n$$

$$\text{LBD} : n_{t+1} = n_t + y_t^n$$

$$g_t^n = \frac{n_{t+1} - n_t}{n_t} = \frac{y_t^n}{n_t}$$

Remark: if $X_t^\alpha = Y_t^\beta$, then

$$\alpha g_x = \beta g_y$$

why?

$$X_t = X_0 (1 + g_x)^t$$

$$Y_t = Y_0 (1 + g_y)^t$$

$$\cancel{X_0 (1 + g_x)^{\alpha t}} = \cancel{Y_0^\beta (1 + g_y)^{\beta t}}$$

$$(1+g_x)^{\alpha t} = (1+g_y)^{\beta t} \quad \forall t > 0$$

Take $t=1$

$$(1+g_x)^{\alpha} = (1+g_y)^{\beta}$$

take logs

$$\log(1+x) \approx x \text{ when } x \approx 0$$

$$\alpha \underbrace{\log(1+g_x)}_{g_x} = \beta \underbrace{\log(1+g_y)}_{g_y}$$

QED

Back to the model:

In the long run g_A and g_n are
constant $\rightarrow$ along a BGP
 $L > a/b \leq 5$

$$\underbrace{g^A}_{\text{growth rate of } g^A = 0} = \underbrace{\pi_1^{\varepsilon-1} (L_t^A)^{\alpha}}$$

$$(\varepsilon-1) g_A + \alpha g_{L^A} = 0$$

Similarly:

$$\underbrace{g^n}_{\text{constant}} = \underbrace{n^{\varepsilon-1} L_t^n}_{\text{constant}}$$

constant

$$(\varepsilon-1)g^n + g_{L^n} = 0$$

— If θ is constant in the long run

then

$$\begin{aligned} L_t^A &= \theta L_t \Rightarrow g_{L_t^A} = g_L \\ L_t^n &= (1-\theta)L_t \quad . \quad g_{L_t^n} = g_L \end{aligned}$$

I have computed :

$$n_t = \frac{\gamma^2 A_t^\varepsilon}{n_t^{d\varepsilon} (\gamma L_t)^{1-\alpha}} = \frac{L_{t+1}}{L_t} = g_{L_t}$$

$$\frac{n_{t+1}}{n_t} = \frac{\gamma^2 A_{t+1}^\varepsilon / n_{t+1}^{d\varepsilon} (\gamma L_{t+1})^{1-\alpha}}{\gamma^2 A_t^\varepsilon / n_t^{d\varepsilon} (\gamma L_t)^{1-\alpha}} = (1+g_{n_{t+1}}) / (1+g_{n_t})^{d\varepsilon} \times n_t$$

$$\Rightarrow \Delta (1 + g_{L,t+1}) = (1 + g_L)^\alpha$$

